

sector, and it performs a variety of tasks.

Pranav Mistry was one of the first few individuals to develop a working prototype of this device, which had such components as camera, projector, camera-sensitive coloured markers, device for interpretation of signals and gestures, plane mirror and sound output device (Fig. 2). The projector and camera were connected to the primary computing device either using external wires or a wireless connection. The device, in turn, was connected to a 4G or 3G network, Wi-Fi network or any other reliable and efficient source of the Internet.

The entire system could be worn by the user around neck. The plane mirror served as an attachment to front part of the projector. The main function of the mirror was to reflect projections of the information on the available surface. The setup was not very expensive and cost around US\$ 350.

The device was given the name Wear Ur World (WUW) due to its design and process of functioning. It involves the principle of gesture recognition to understand the gestures captured by the input device with the help of computer visualisation algorithms.

Nowadays, WUW is simply referred to as Sixth Sense technology. Pranav Mistry has published a beta software version of this device written in C++ and C# languages, which is a part of Google Code project, and has also commissioned project name WUW.

According to Mistry, Sixth Sense is a wearable gestural interface that digitally alters the physical world and lets the user use natural hand signals and indications to interact with digital information. Using a camera and a small projector affixed atop support, Sixth Sense serves as the user's eyes and visually alters any surface or object the user is interacting with, and uses it for displaying output. Sixth Sense aims

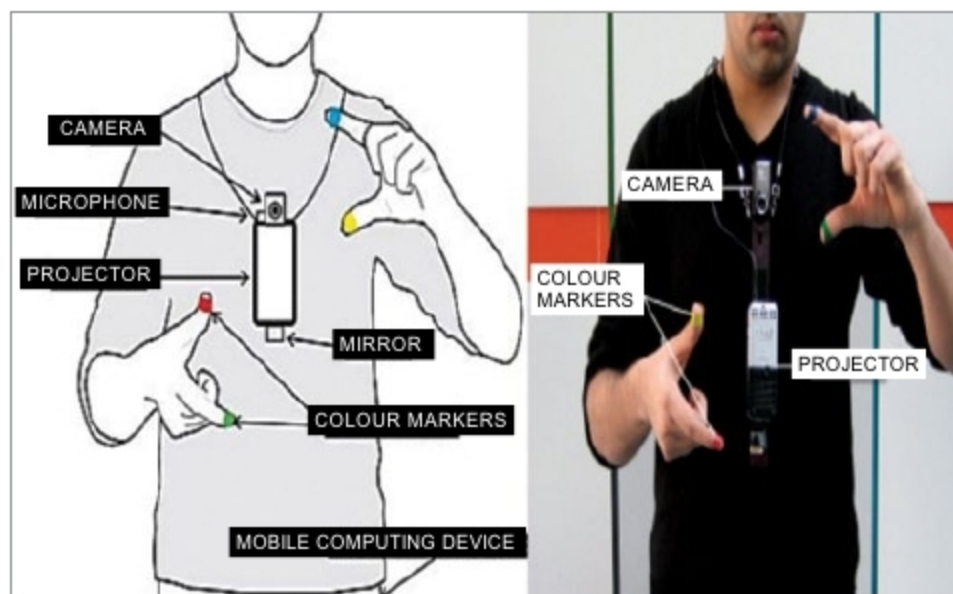


Fig. 2: Pranav Mistry with Sixth Sense technology prototype (Credit: www.pranavmistry.com)

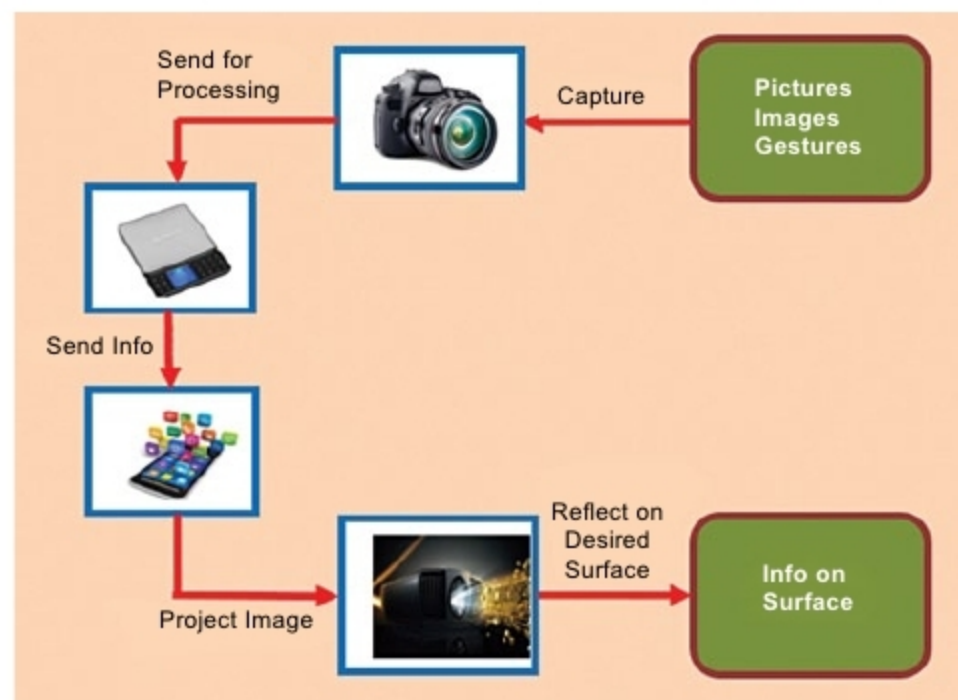


Fig. 3: Working of Sixth Sense technology

at providing an unprecedented level of freedom in the sphere of technology.

Working mechanism of this device involves multiple components working in unison. The camera serves as the input device for the gestures made by the user's hand. The user's fingertips are enveloped by different colour markers that are sensitive to the camera.

The camera consists of sensors to understand the motion and dynamics of gestures. Images of objects and gestures, which have been streamed, are identified as commands by the computing device and interpreted using the principles of coordinates and location of objects on the screen.

The device then works as per instructions and may send queries to the Internet, if needed. It then returns the information required by the user. This information may be projected via the output device while the plane

mirror reflects it onto the desired surface or object. These commands may be followed by another instruction.

Concepts and technology behind the device

Different technological concepts are implemented through WUW. Some of these include ubiquitous computing, AR, gesture recognition, computer vision and radio frequency identification (RFID).

Ubiquitous computing is a form of computing that involves greater use of computers and technological devices to such an extent that they start affecting all actions and tasks performed by individuals. Sixth Sense technology may pave the way for this form of technology.

Developing an ubiquitous computing framework, while maintaining confidentiality of consumers at the same time, is indeed a monumental task for engineers of today. WUW device lives up to this principle as

it provides flawless information about various different subjects.

AR involves interaction of digital information and visual displays with the physical world through phones and smart devices. This concept is implemented by the device by integrating the real-world environment with information and visuals.

The process of recognising human gestures is known as signal or gesture recognition, and the computer interface using this method is called gesture-recognition technology. This technology is a much better and more convenient substitute for text and graphical user interfaces as it greatly saves time and provides greater levels of freedom and interactions.

Applications and uses

WUW has a very wide range of applications that are especially relevant in today's world. Some of these are: